Harshit Hirpara

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.83
B.H Rathod Higher Secondary School	Class XII (Gujarat Secondary & Higher Secondary Education Board)	2024	91.69%
Kendriya Vidyalaya Rajkot	Class X (CBSE)	2022	96.20%

SCHOLASTIC AND TECHNICAL ACHIEVEMENTS

- **Grand Finalist (National top 20) Integral Cup 2025** conducted by **Optiver, QRT, Jane Street.**
- **ICPC India National Preliminaries (2025):** Secured Overall **Rank 122** out of 4,000+ competing teams.
- Achieved a Maximum Rating of **1718 (Expert)** on Codeforces (profile) , solved **1200+** problems across all platforms.
- **3-star Coder** on Codechef (profile) with peak rating of **1754**, achieved **Global Rank 54** in **Starters 245.**
- Achieved **Global rank 38** and **All India Rank 8** in **Codeforces Round 1103 (Div. 3)** among 41,000+ participants.
- **State Topper** of **Gujarat** and **Top 1%** in **NSEC (IAPT) 2023** .
- Ranked in the **Top 1% in NSEP (IAPT) 2023** among 49,000+ candidates.
- Awarded **Bronze Medal** in **SASMO (Singapore and Asian Schools Math Olympiad) 2022.**

PROJECTS

xv6 OS Kernel Extension | *C, RISC-V, Systems Programming* | 2026

- Built a concurrency-safe virtual memory subsystem for xv6-riscv with eager/lazy sbrk, trap-driven demand paging, Clock page replacement, and disk-backed swapping beyond physical RAM.
- Replaced the default scheduler with a custom 4-level System-Call-Aware MLFQ using exponential quanta, syscall-aware prioritization, and periodic boosts to improve responsiveness and prevent starvation.
- Developed a disk-backed swap system with FCFS/SSTF disk scheduling and support for RAID 0, RAID 1, and RAID 5 storage abstractions.
- Eliminated multi-core page-eviction races using spinlocks and strict lock ordering; validated stability with 32 processes on 8 SMP cores, 1,700+ page evictions, and zero kernel panics or data corruption.

Advanced Pipelined RISC-V Processor Simulator | *C++, Computer Architecture,CMake* | 2025

- Built a cycle-accurate 5-stage pipelined RISC-V processor simulator from a single-cycle design.
- Implemented hazard detection, data forwarding, and stalling to handle pipeline dependencies.
- Developed static and 1-bit dynamic branch prediction to reduce branch penalties.
- Designed a configurable L1 cache and benchmarked CPI, pipeline performance, and cache hit rates using assembly workloads.

Core-ML Exception Handling Extension | *PL Zoo* | 2025

- Extended the miniml language syntax and semantics to fully support try-catch exception handling.
- Implemented IRaise and IHandle structures across the compiler, abstract machine, and type checker.
- Developed robust division operations featuring dedicated division-by-zero handling and comprehensive test cases.

Arbitrary Precision Calculator | *Java* | 2025

- Developed a Java-based calculator capable of processing arbitrary-size integers and floating-point numbers.
- Implemented core arithmetic operations including addition, subtraction, multiplication, and division.
- Enabled precise computations that successfully bypass the limitations of native data types.

SKILLS

- **Languages:** C, C++, Python, Java, OCaml, Assembly, Bash,SQL,HTML,CSS,Javascript
- **Tools:** Git/GitHub, LaTeX, Solid Edge,PostgreSQL
- **Soft Skills:** Communication, Work Ethic, Time Management, Teamwork, Problem Solving

RELEVANT COURSES

- **CS** - Intro to Programming, Software Development Fundamentals, Algorithms, Computer Architecture, Operating System I & II, Database management system I&II , Data Structure and Algorithms, Theory of computation, Compilers-I,
- **Mathematics** - Calculus-I, Calculus-II, Discrete Mathematics for Computer Science, Probability and Random Variables, Transform Techniques, Linear Algebra, Intro to statisticcs, Applied Statistics

POSITIONS OF RESPONSIBILITY

Senior Core Member, Infero

Algorithmic and Logical Reasoning Club, IIT Hyderabad

- Guided core members and juniors in their competitive programming journey by mentoring them.

Core Member, Infero (*Algorithmic and Logical Reasoning Club, IIT Hyderabad)*

- Contributed to problem-setting for Elan&Nvision Codeclash 2025.
- Mentored juniors in competitive programming through sessions and contests.

Teaching Assistant, *Introduction to Programming (ID1063)*

EXTRACURRICULAR

- **Co-ordinator** Of NvisionX (Techfest club of IITH)
- **Member** of RDM (Drama club of IITH)